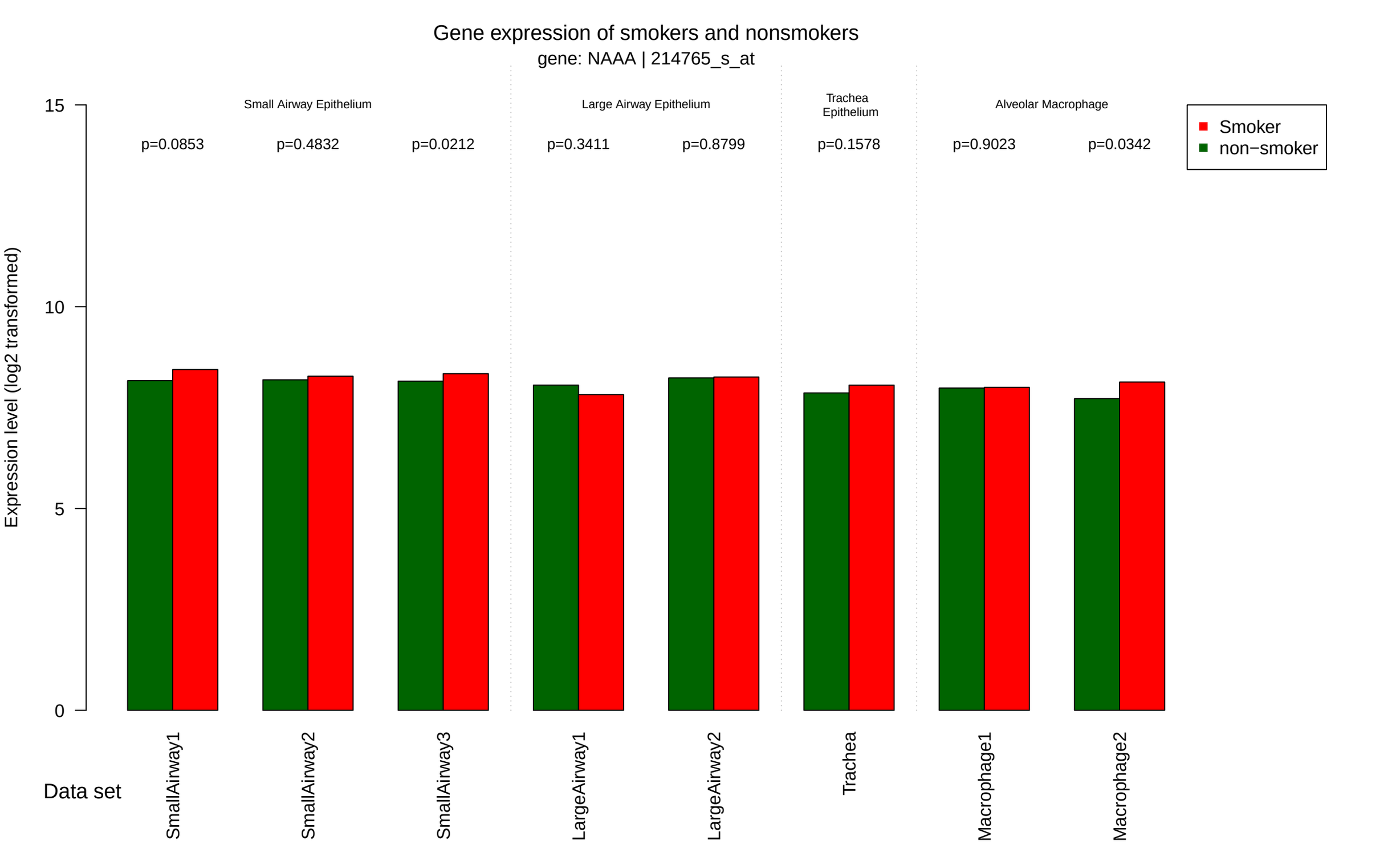
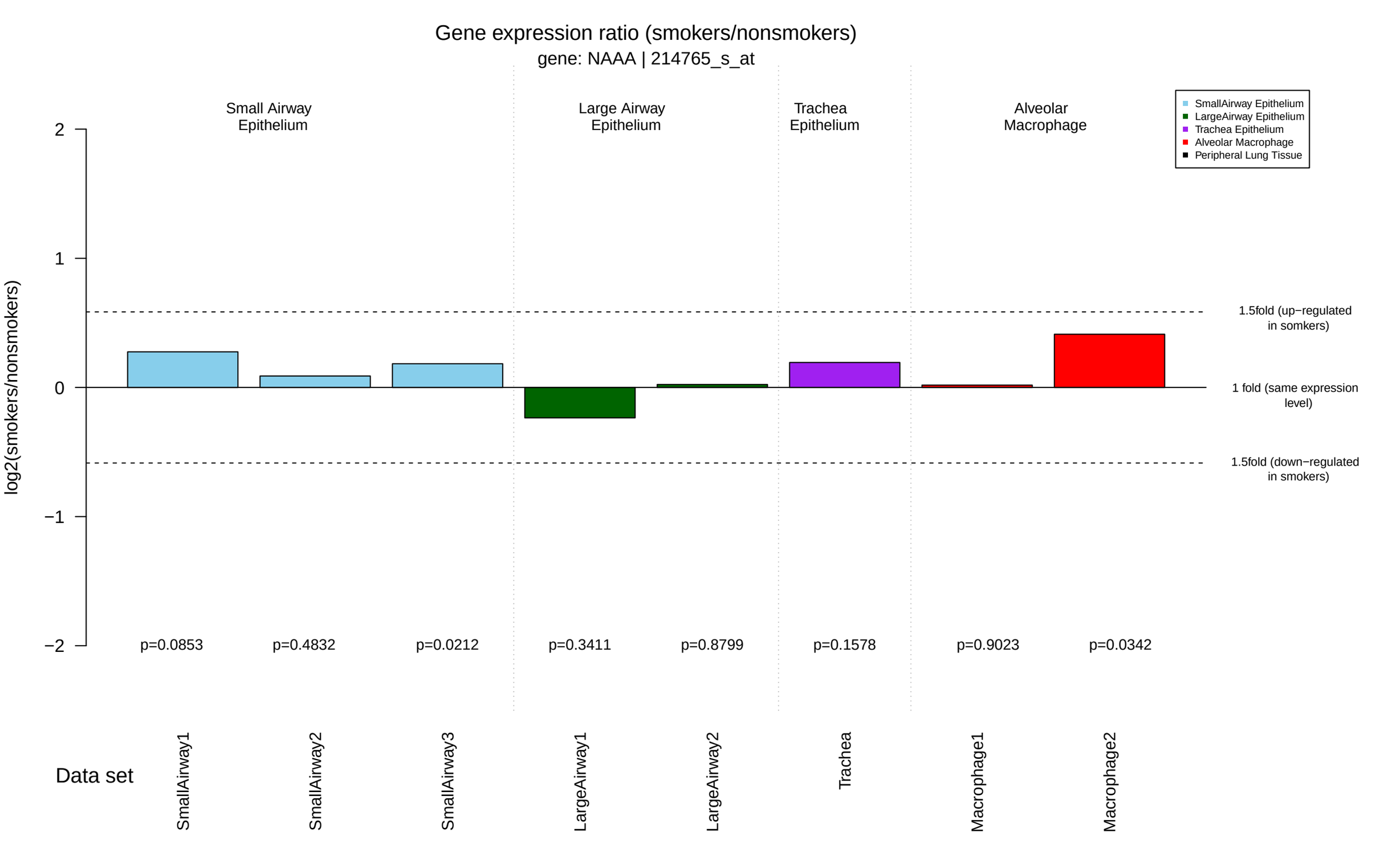
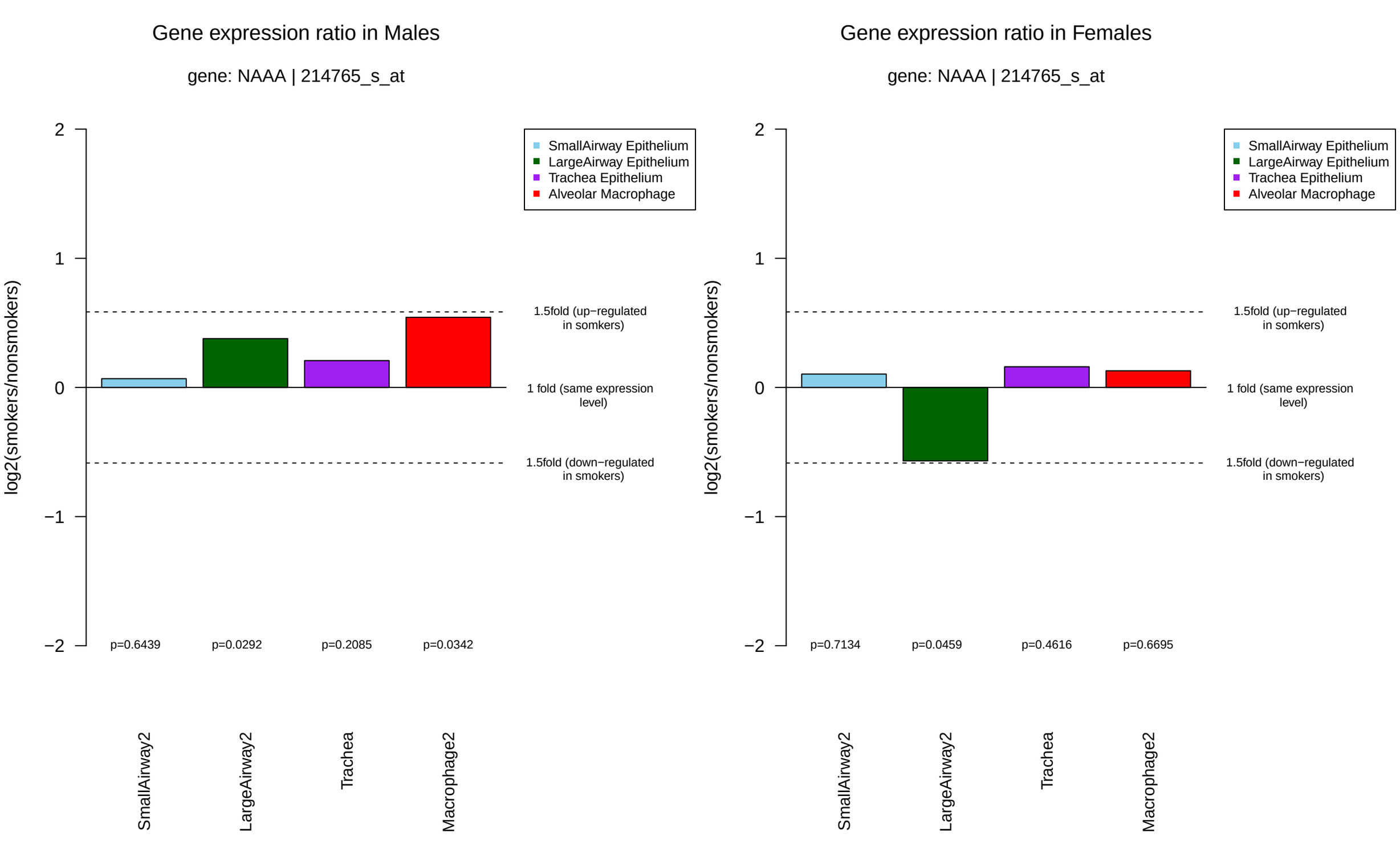
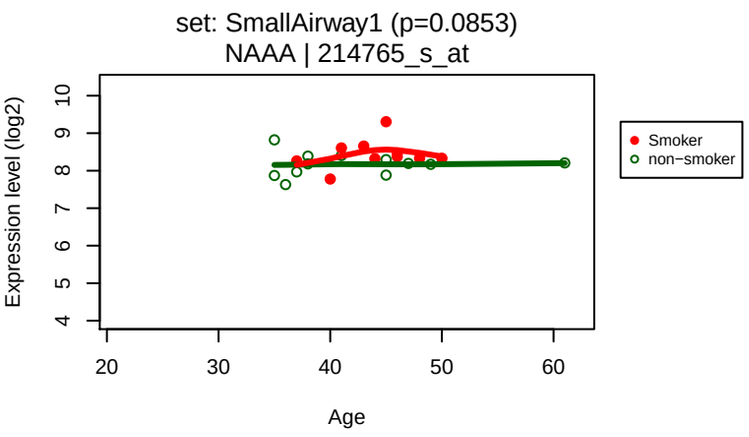
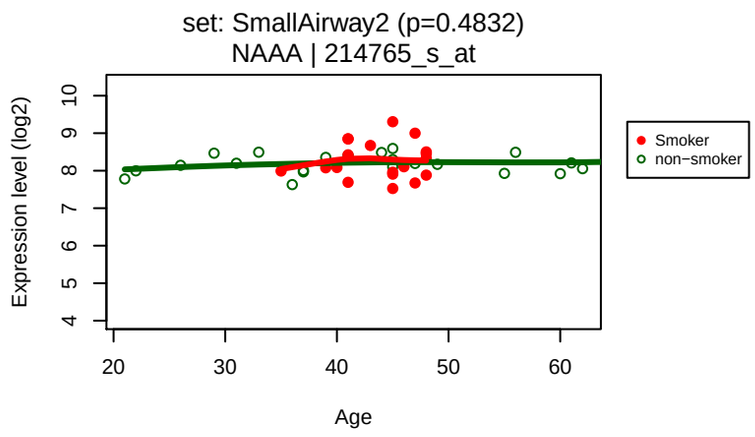
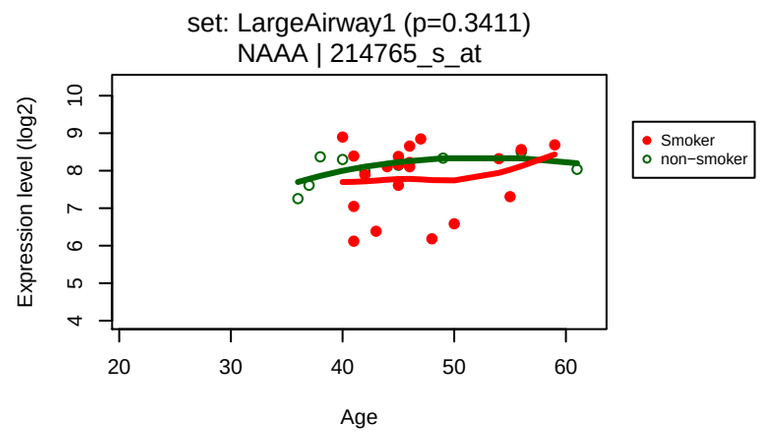

Gene expression of smokers and nonsmokers
gene: NAAA | 214765_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0853
p=0.4832
p=0.0212
p=0.3411
p=0.8799
p=0.1578
p=0.9023
p=0.0342
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

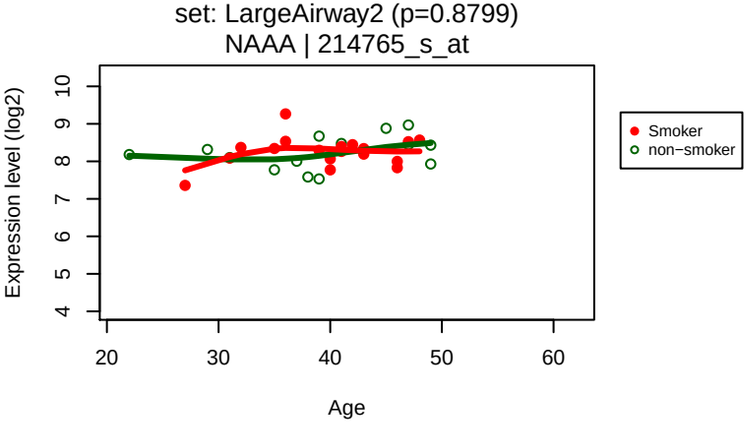

Gene expression ratio (smokers/nonsmokers)
gene: NAAA | 214765_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0853
p=0.4832
p=0.0212
p=0.3411
p=0.8799
p=0.1578
p=0.9023
p=0.0342
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

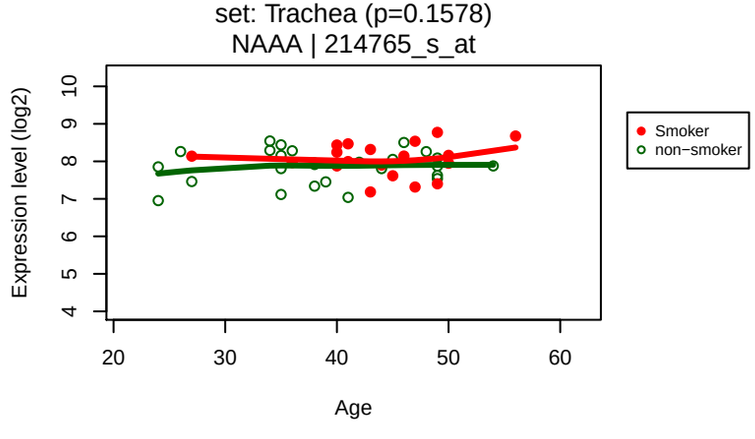

Gene expression ratio in Males
gene: NAAA | 214765_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6439
p=0.0292
p=0.2085
p=0.0342
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NAAA | 214765_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7134
p=0.0459
p=0.4616
p=0.6695
SmallAirway2
LargeAirway2
Trachea
Macrophage2

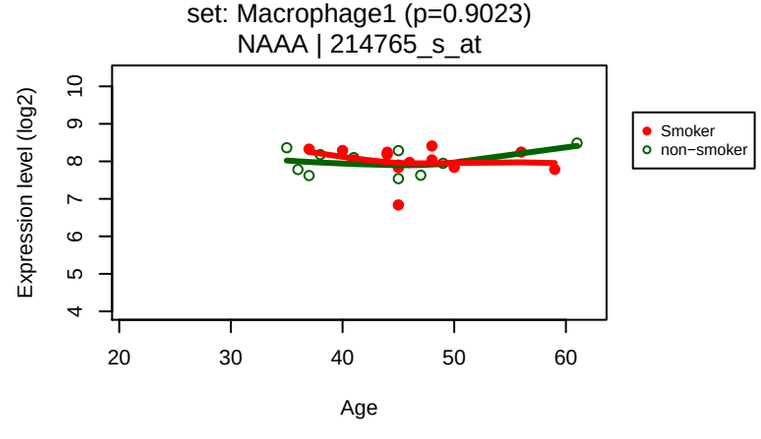

set: SmallAirway1 (p=0.0853)
NAAA | 214765_s_at
Expression level (log2)
Age
Smoker
non−smoker

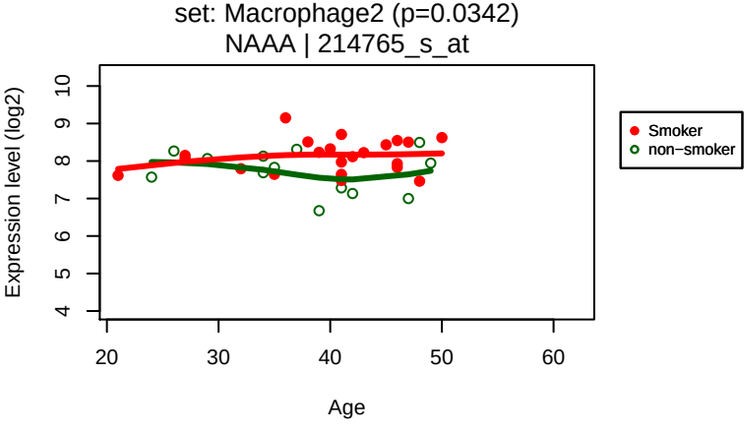

set: SmallAirway2 (p=0.4832)
NAAA | 214765_s_at
Expression level (log2)
Age
Smoker
non−smoker

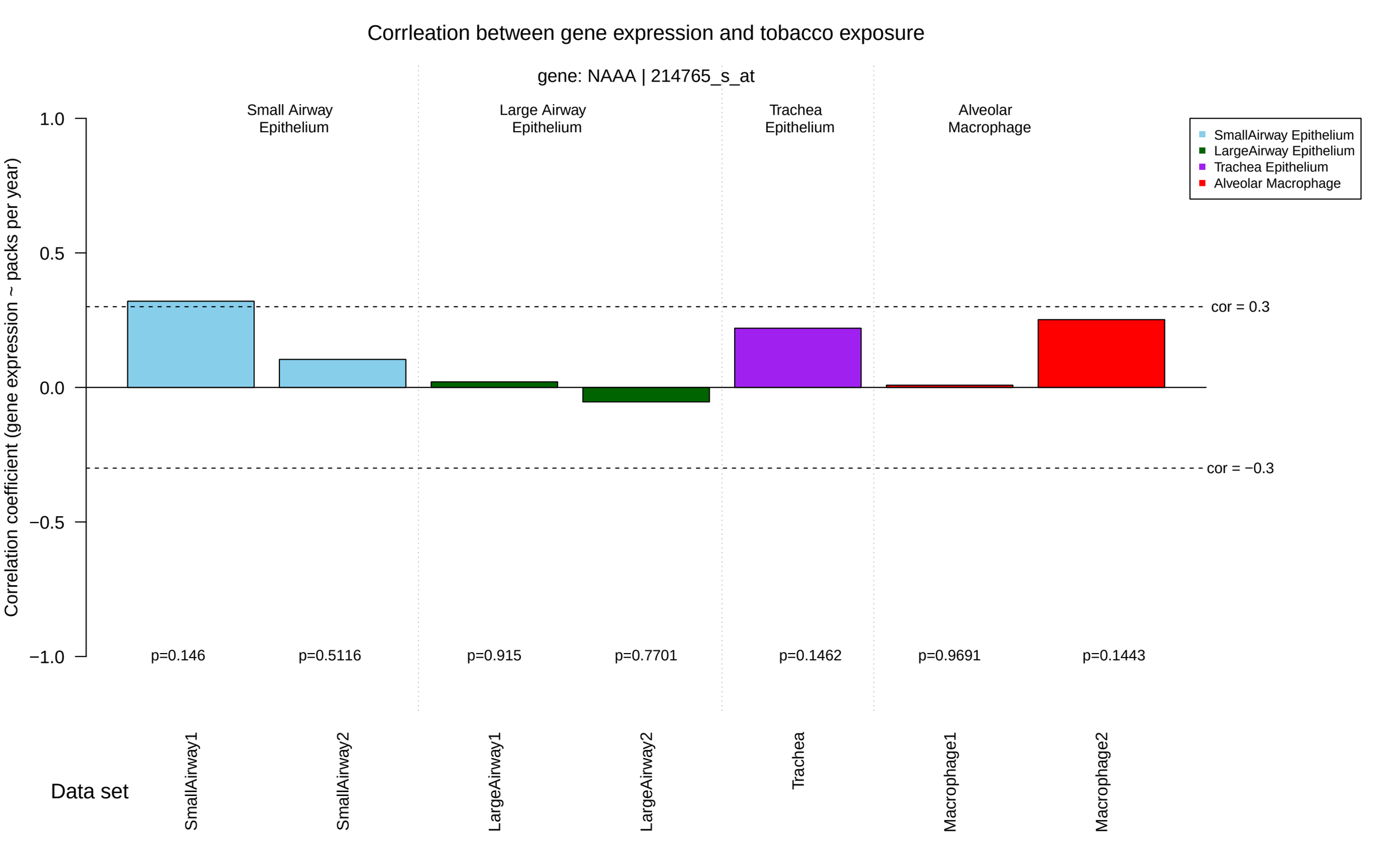

set: LargeAirway1 (p=0.3411)
NAAA | 214765_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8799)
NAAA | 214765_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1578)
NAAA | 214765_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9023)
NAAA | 214765_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0342)
NAAA | 214765_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NAAA | 214765_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.146
p=0.5116
p=0.915
p=0.7701
p=0.1462
p=0.9691
p=0.1443
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2